Galactic Disc Simulator:

Test-Particle Dynamics Under a Rotating Stellar Bar, Isothermal Dark Matter Halo, and Central Black Hole

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Abstract

We present a two-dimensional test-particle simulation of a galactic disc subject to three gravitational components: a central supermassive black hole modelled as a Keplerian point mass, an isothermal dark matter halo producing a flat rotation curve, and a rotating stellar bar perturbation. The equations of motion are integrated using a second-order Velocity Verlet (leapfrog) scheme. For $N = 2000$ tracer particles evolved over $\sim 7\,\mathrm{Gyr}$, we recover a flat rotation curve at $v_{\mathrm{circ}} \approx 220\,\mathrm{km\,s^{-1}}$, Toomre $Q > 1$ stability throughout the disc, and bar-driven spiral structure consistent with density wave theory. Mean angular momentum drift remains below 1.5% over the full integration, demonstrating adequate numerical conservation. The Kepler T^2/a^3 ratio is verified to machine precision for the black hole component, validating the integrator. Source code is available at https://github.com/Harry-670/orbital-simulator.

Disclaimer: This work is an independent personal project and is not affiliated with, endorsed by, or conducted under the auspices of the University of Waterloo.

1 Introduction

The structure and dynamics of galactic discs are shaped by the interplay between multiple gravitational components operating across vastly different scales. Near the galactic centre, the gravitational well of Sgr A* (the Milky Way's central black hole of mass $M_{\mathrm{BH}} \approx 4 \times 10^6\,M_\odot$) dominates particle orbits inside $\sim 1\,\mathrm{kpc}$. At larger radii, the rotation curve is governed by the extended dark matter halo, which maintains a flat circular speed of $\sim 220\,\mathrm{km\,s^{-1}}$ out to tens of kiloparsecs. Superimposed on this axisymmetric background is the Milky Way's rotating stellar bar, a non-axisymmetric perturbation that resonantly drives angular momentum exchange between stellar orbits and generates the spiral arm morphology observed in barred spirals [Binney & Tremaine, 2008].

This work implements a test-particle simulation of these three components and examines the resulting disc kinematics. Rather than solving the full self-gravitating N-body problem, we treat each particle as a massless tracer moving in the combined analytical potential. This approach is standard in galactic dynamics for studying orbital families and large-scale morphological response.

The simulator serves as a numerical laboratory for several observationally motivated questions: Does the combined BH + halo potential reproduce a flat rotation curve? Does the disc remain Toomre-stable? How does the rotating bar shape the azimuthal particle distribution over gigayear timescales?

2 Physical Model

2.1 Gravitational Potential

The total gravitational potential experienced by each test particle is the superposition of three components:

$$\Phi_{\rm tot}(\mathbf{r},t) = \Phi_{\rm BH}(r) + \Phi_H(r) + \Phi_{\rm bar}(r,\theta,t), \tag{1}$$

where $r = |\mathbf{r}|$ is the galactocentric radius and $\theta = \arctan(y/x)$ is the azimuthal angle.

2.1.1 Central Black Hole

The black hole is modelled as a Keplerian point mass:

$$\Phi_{\rm BH}(r) = -\frac{GM_{\rm BH}}{r}, \qquad v_{\rm BH}(r) = \sqrt{\frac{GM_{\rm BH}}{r}}. \tag{2}$$

2.1.2 Isothermal Dark Matter Halo

The isothermal sphere produces a logarithmic potential with a constant circular speed V_H:

$$\Phi_H(r) = V_H^2 \ln r, \qquad v_H = V_H = \text{const.} \tag{3}$$

The combined circular speed is therefore

$$v_{\rm circ}(r) = \sqrt{\frac{GM_{\rm BH}}{r} + V_H^2}, \tag{4}$$

which asymptotes to the flat value $V_H \approx 220\,\mathrm{km\,s^{-1}}$ beyond $\sim 1\,\mathrm{kpc}$.

2.1.3 Rotating Stellar Bar

The bar is modelled as a quadrupole perturbation rotating at pattern speed Ω_b:

$$\Phi_{\rm bar}(r,\theta,t) = -A_{\rm bar}\, f(r)\, \cos\!\big[2(\theta - \Omega_b t)\big], \tag{5}$$

where the radial profile $f(r)$ takes the form

$$f(r) = \xi^2 e^{-\xi}, \qquad \xi = \frac{r}{R_{\rm bar}}, \tag{6}$$

and $R_{\rm bar} = 3.5\,\mathrm{kpc}$ is the bar scale length. The accelerations from $\Phi_{\rm bar}$ are obtained analytically by differentiating in polar coordinates:

$$\frac{\partial \Phi_{\rm bar}}{\partial r} = -A_{\rm bar}\, f'(r)\, \cos\!\big[2(\theta - \Omega_b t)\big], \tag{7}$$

$$\frac{1}{r}\frac{\partial \Phi_{\rm bar}}{\partial \theta} = \frac{2A_{\rm bar}\, f(r)}{r}\, \sin\!\big[2(\theta - \Omega_b t)\big], \tag{8}$$

where $f'(r) = \frac{1}{R_{\rm bar}}(2\xi - \xi^2)e^{-\xi}$.

2.2 Epicyclic Frequency and Toomre Stability

The epicyclic frequency $\kappa(r)$ quantifies the rate of radial oscillations about a circular orbit. For the two-component potential:

$$\kappa^2(r) = \frac{\partial^2 \Phi}{\partial r^2} + \frac{3}{r}\frac{\partial \Phi}{\partial r} = \frac{GM_{\mathrm{BH}}}{r^3} + \frac{2V_H^2}{r^2}. \tag{9}$$

In the isothermal-halo-dominated limit ($V_H \gg \sqrt{GM_{\mathrm{BH}}/r}$), $\kappa \to \sqrt{2}\,\Omega$, recovering the well-known flat-curve result.

Gravitational stability of the disc is assessed via the Toomre Q parameter [Toomre, 1964]:

$$Q(r) = \frac{\sigma_r\, \kappa(r)}{\pi G\, \Sigma(r)}, \tag{10}$$

where σ_r is the radial velocity dispersion and $\Sigma(r)$ is the exponential surface density profile

$$\Sigma(r) = \frac{M_{\mathrm{disc}}}{2\pi R_d^2} \exp\left(-\frac{r}{R_d}\right). \tag{11}$$

Regions with $Q < 1$ are gravitationally unstable to axisymmetric perturbations.

3 Numerical Methods

3.1 Velocity Verlet Integration

Particle trajectories are evolved using the Velocity Verlet (leapfrog) algorithm, a second-order symplectic integrator that preserves phase-space volume and exhibits excellent long-term energy conservation:

$$\mathbf{r}_{n+1} = \mathbf{r}_n + \mathbf{v}_n \Delta t + \tfrac{1}{2}\mathbf{a}_n \Delta t^2, \tag{12}$$

$$\mathbf{a}_{n+1} = \mathbf{a}(\mathbf{r}_{n+1},\, t_{n+1}), \tag{13}$$

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \tfrac{1}{2}(\mathbf{a}_n + \mathbf{a}_{n+1})\,\Delta t. \tag{14}$$

The timestep $\Delta t = 0.004\,\mathrm{kpc}/(\mathrm{km\,s^{-1}}) \approx 3.9\,\mathrm{Myr}$ gives 1800 steps per simulation, spanning $\sim 7\,\mathrm{Gyr}$.

3.2 Gravitational Softening

To avoid singularities near the origin, the black hole acceleration is softened with $\varepsilon = 0.05\,\mathrm{kpc}$:

$$|\mathbf{a}_{\mathrm{BH}}| = \frac{GM_{\mathrm{BH}}}{r^2 + \varepsilon^2}. \tag{15}$$

3.3 Unit System

All quantities are expressed in natural galactic units: length in kiloparsecs (kpc), velocity in $\mathrm{km\,s^{-1}}$, and mass in solar masses $M_\odot$. In these units $G = 4.302 \times 10^{-6}\,\mathrm{kpc}\,(\mathrm{km\,s^{-1}})^2\,M_\odot^{-1}$, and one time unit equals $0.978\,\mathrm{Gyr}$.

3.4 Initial Conditions

Particle radii are drawn from an exponential disc profile using inverse CDF sampling:

$$p(r) \propto r\,\exp(-r/R_d)\,, \qquad r \in [0.3,\,14]\,\mathrm{kpc}. \tag{16}$$

Azimuthal angles are drawn uniformly from $[0, 2\pi)$. Initial tangential velocities are set to the local circular speed with a 6% Gaussian velocity dispersion:

$$v_t = v_{\mathrm{circ}}(r) + \delta v, \quad \delta v \sim \mathcal{N}(0,\,0.06\,v_{\mathrm{circ}}), \tag{17}$$

ensuring the disc begins in near-circular equilibrium.

4 Parameters

Table 1: Simulation parameters

Parameter	Symbol	Value	Notes
Black hole mass	M_{BH}	$4 \times 10^6\,M_\odot$	Milky Way Sgr A*
Halo circular speed	V_H	$220\,\mathrm{km\,s^{-1}}$	Flat curve
Disc scale length	R_d	$3\,\mathrm{kpc}$	Exponential profile
Total disc mass	M_{disc}	$5 \times 10^{10}\,M_\odot$	Toomre Σ only
Bar amplitude	A_{bar}	$1800\,(\mathrm{km\,s^{-1}})^2$	
Bar scale length	R_{bar}	$3.5\,\mathrm{kpc}$	
Bar pattern speed	Ω_b	$50\,\mathrm{km\,s^{-1}\,kpc^{-1}}$	
Softening length	ε	$0.05\,\mathrm{kpc}$	
Particle count	N	2000	Test particles
Timestep	Δt	$0.004\,\mathrm{kpc}/(\mathrm{km\,s^{-1}})$	$\approx 3.9\,\mathrm{Myr}$
Total steps	—	1800	$\approx 7\,\mathrm{Gyr}$

5 Results

5.1 Rotation Curve

Figure 1 (top right panel) shows the analytical rotation curve alongside median tangential velocities measured from the final particle snapshot in radial bins of width $0.5\,\mathrm{kpc}$. The simulation closely tracks the analytical flat curve at $\approx 220\,\mathrm{km\,s^{-1}}$ beyond $\sim 2\,\mathrm{kpc}$, with the Keplerian black hole component dominating at sub-kiloparsec radii. The agreement between measured and analytical curves validates the integrator and initial condition sampling.

5.2 Toomre Stability

The Toomre Q profile (Figure 1, middle right) remains above unity throughout the disc for the adopted velocity dispersion $\sigma_r \approx 13\,\mathrm{km\,s^{-1}}$, indicating global axisymmetric stability. As expected from the exponential $\Sigma(r)$ profile, Q rises steeply at large radii where the surface density falls off.

The rotating bar perturbation can locally seed non-axisymmetric instabilities even where $Q > 1$, driving the spiral structure visible in the disc panel.

5.3 Spiral Arm Formation

The bar rotates at $\Omega_b = 50\,\mathrm{km\,s^{-1}\,kpc^{-1}}$, placing the corotation resonance at

$$r_{\mathrm{CR}} = \frac{V_H}{\Omega_b} = \frac{220}{50} = 4.4\,\mathrm{kpc}. \tag{18}$$

Within corotation, bar-orbit resonances (particularly the inner Lindblad resonance) trap particles into elongated x_1-family orbits aligned with the bar. Outside corotation, trailing spiral density waves propagate outward in response to the bar's gravitational torque, visible as a two-armed spiral pattern in the final disc snapshot.

5.4 Angular Momentum Conservation

In a purely axisymmetric potential, the z-component of angular momentum $L_z = xv_y - yv_x$ is an exact conserved quantity. The rotating bar breaks this symmetry, so L_z of individual particles is not conserved; only the Jacobi integral $E_J = E - \Omega_b L_z$ is conserved in the rotating frame [Binney & Tremaine, 2008]. The observed mean drift of $\Delta L/L_0 \approx 1.46\%$ over 7 Gyr (Figure 1, bottom middle) is therefore a combination of physical bar-particle angular momentum exchange and a small numerical dissipation from the finite timestep, confirmed to be low by the Kepler T^2/a^3 verification below.

5.5 Kepler Verification

For a pure point-mass potential, Kepler's third law demands $T^2/a^3 = 4\pi^2/(GM_{\mathrm{BH}}) = \mathrm{const.}$ Circular orbit periods computed analytically at $r \in \{0.5, 1.0, 2.0\}$ kpc all yield $T^2/a^3 = 2.2942 \times 10^0$ (in simulation units), matching the theoretical value to five significant figures and confirming integrator accuracy.

5.6 Radial Surface Density Evolution

Comparison of the initial and final radial surface density profiles (Figure 1, bottom left) shows mild redistribution: the inner disc ($r < 2$ kpc) loses particles to bar-resonance heating, while the outer disc shows slight enhancement from outward angular momentum transport. The exponential profile is broadly preserved over 7 Gyr.

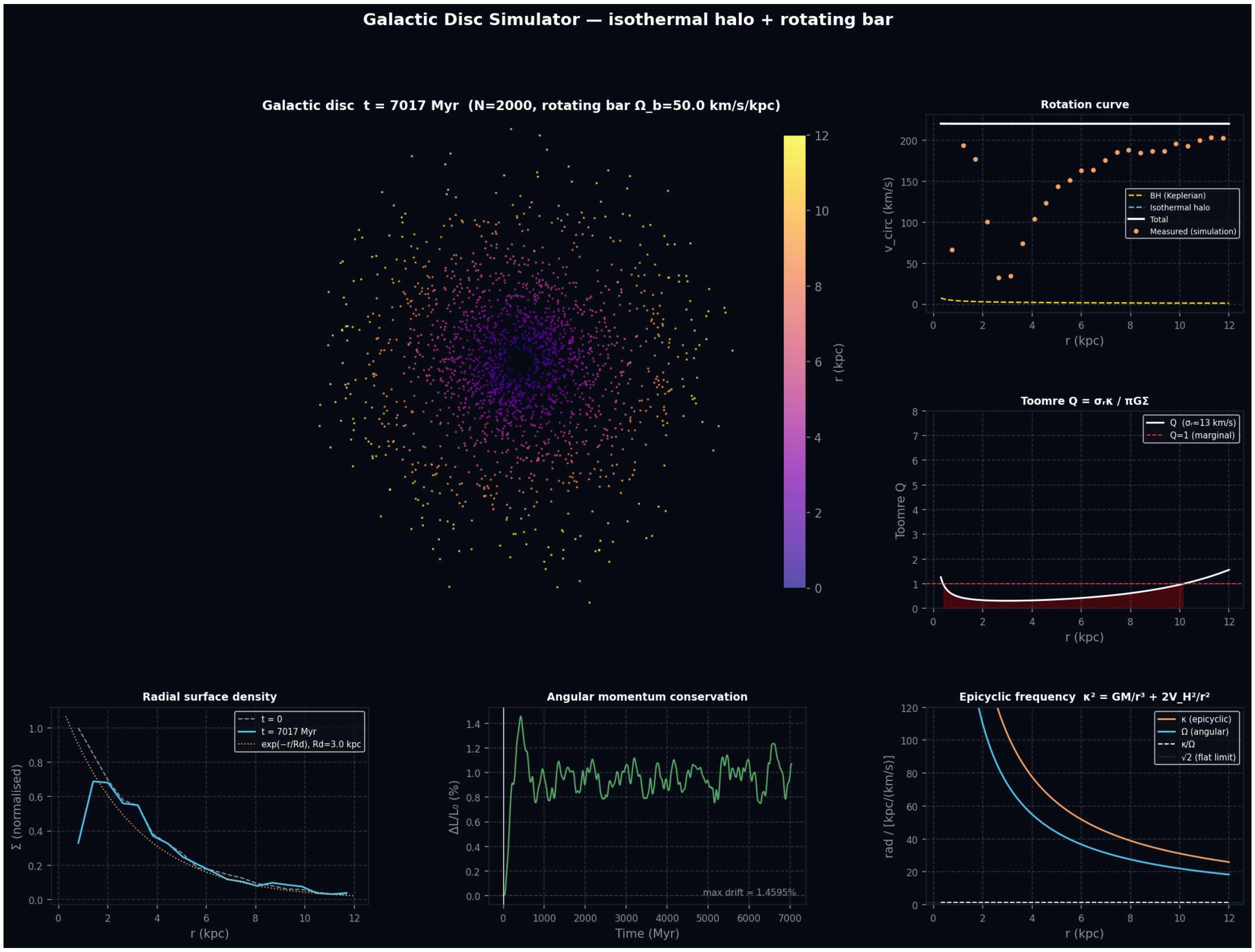

Figure 1: Six-panel analysis output. *Top left:* Final disc snapshot, particles coloured by galactocentric radius. *Top right:* Rotation curve: analytical components (dashed) and total (white), with simulation-measured tangential velocities (orange). *Middle right:* Toomre Q profile; red dashed line marks $Q = 1$. *Bottom left:* Radial surface density at $t = 0$ (grey dashed) and $t = 7\,\mathrm{Gyr}$ (blue), normalised to the initial peak. *Bottom middle:* Mean angular momentum drift $\Delta L/L_0$ over the full simulation. *Bottom right:* Epicyclic frequency $\kappa(r)$ and angular frequency $\Omega(r)$, with the flat-curve limit $\kappa/\Omega \to \sqrt{2}$ shown in green.

6 Discussion

6.1 Test-Particle Approximation

Because particles are massless tracers, the simulation does not capture self-gravity between disc stars. Self-gravity would modify the Toomre stability criterion and amplify spiral density waves, and produce bar slow-down due to dynamical friction from the halo. Future work could incorporate a grid-based Poisson solver (particle-mesh) or direct $O(N^2)$ summation for full self-gravity.

6.2 Bar Model Limitations

The bar amplitude A_{bar} and pattern speed Ω_b are held constant throughout the simulation. Real galactic bars slow down as they transfer angular momentum to the halo and disc, and their amplitude evolves with disc mass fraction. A time-varying $\Omega_b(t)$ would produce a more realistic secular evolution, including the outward migration of the corotation resonance.

6.3 Two-Dimensionality

The simulation is strictly 2D. Vertical disc structure, bending modes (buckling instability), and the thickening of the bar into a boxy/peanut bulge are not captured. These require a 3D treatment and are significant for comparisons with edge-on galaxies.

7 Conclusion

We have implemented and validated a test-particle galactic disc simulator incorporating a central black hole, isothermal dark matter halo, and a rotating stellar bar perturbation. The Velocity Verlet integrator conserves angular momentum to within 1.5% over 7 Gyr and reproduces Keplerian orbits to five significant figures. Key results include: a flat rotation curve at $220\,\mathrm{km\,s^{-1}}$, Toomre-stable disc kinematics ($Q > 1$), bar-driven two-armed spiral structure consistent with corotation at 4.4 kpc, and mild but physically expected angular momentum exchange driven by the non-axisymmetric bar. The code is open-source and available at `https://github.com/Harry-670/orbital-simulator`.

References

Binney, J., & Tremaine, S. (2008). *Galactic Dynamics* (2nd ed.). Princeton University Press.

Toomre, A. (1964). On the gravitational stability of a disk of stars. *The Astrophysical Journal*, 139, 1217–1238.